Functional Annotation of *pycA* (PP_5347; UniProt Q88C36) in *Pseudomonas putida* KT2440

Summary

pycA (locus PP_5347; UniProt Q88C36) in *Pseudomonas putida* KT2440 encodes the biotin carboxylase (BC), or α, subunit of a two-subunit pyruvate carboxylase (EC 6.4.1.1). Its primary molecular function is to catalyze the first, ATP- and bicarbonate-dependent half-reaction of biotin-dependent carboxylation: it carboxylates the biotin prosthetic group that is covalently attached to a lysine of its partner subunit, PycB. Working together, PycA and PycB convert pyruvate + HCO_3^- + ATP into oxaloacetate + ADP + P_i. This is a soluble, cytoplasmic **anaplerotic** reaction — it replenishes the four-carbon dicarboxylic-acid pool of the tricarboxylic acid (TCA) cycle and supplies oxaloacetate for gluconeogenesis and biosynthesis.

An important correction emerged during this investigation. UniProt's automated (ARBA/rule-based) annotation labels Q88C36 as the "acetyl-coenzyme A carboxylase biotin carboxylase subunit A." This name is derived from the shared ATP-grasp/biotin-carboxylase domain architecture that is common to **all** biotin-dependent carboxylases, and it is a **mis-annotation** in the specific genomic context of *P. putida* KT2440. Genomic synteny, orthology (KEGG), and domain analysis all indicate that PP_5347 is the BC subunit of pyruvate carboxylase, not of acetyl-CoA carboxylase. Decisively, *P. putida* KT2440 possesses a *separate, complete* acetyl-CoA carboxylase encoded by *accBCAD* (accC = PP_0558, accB = PP_0559, accA = PP_1607, accD = PP_1996), so PycA is neither required for nor part of fatty-acid initiation.

The evidence base combines (i) direct genomic and orthology evidence within *P. putida* KT2440, (ii) sequence and structural bioinformatics on Q88C36 itself (51% identity to *E. coli* biotin carboxylase with a fully conserved active site; soluble cytoplasmic protein with no signal peptide or transmembrane helix), and (iii) a rich mechanistic literature on the biotin-dependent carboxylase superfamily and, specifically, the distinct two-subunit pyruvate carboxylase architecture found in Gram-negative bacteria. Multi-omics data from *P. putida* KT2440 grown on lignin-derived aromatic substrates provide physiological confirmation that the enzyme is upregulated for anaplerotic TCA replenishment.

Key Findings

Finding 1 — pycA is the biotin carboxylase subunit of a two-subunit pyruvate carboxylase, not of acetyl-CoA carboxylase

The single most consequential result of this investigation is a **correction to the database annotation.** The initial reading of UniProt Q88C36 ("Biotin carboxylase"; alt. name "Acetyl-coenzyme A carboxylase biotin carboxylase subunit A") suggested a role in fatty-acid initiation. However, examination of the genomic context, orthology assignments, and domain content of the neighboring gene overturns this.

Several independent lines of evidence converge:

1. **Genomic synteny.** PP_5347 (*pycA*) sits immediately adjacent to PP_5346 (*pycB*, UniProt Q88C37, 602 aa), which KEGG annotates as **pyruvate carboxylase subunit B (K01960)**. Adjacent, co-transcribed α/β genes are the hallmark of the two-subunit pyruvate carboxylase.
2. **Domain partitioning.** PycB contains a **pyruvate carboxyltransferase (CT) domain (IPR000891/PF00682)** plus a C-terminal **biotin/lipoyl-attachment (BCCP) domain** (residues ~526–601) — i.e., PycB carries the biotin. PycA (471 aa) contains **only** the biotin-carboxylase/ATP-grasp domain and has **no** BCCP domain. This division of labor is exactly the two-subunit PC architecture.
3. **Orthology.** KEGG assigns PP_5347 → **K01959 (pyruvate carboxylase subunit A, EC 6.4.1.1)**, mapped to the TCA cycle and pyruvate metabolism — not to fatty-acid biosynthesis.
4. **A separate ACC exists.** *P. putida* KT2440 encodes a complete, dedicated acetyl-CoA carboxylase elsewhere in the genome: accC = PP_0558 (K01961), accB = PP_0559 (K02160), accA = PP_1607, accD = PP_1996. Because a full ACC is already present, PP_5347 is not needed for malonyl-CoA production.

This architecture matches the described two-subunit pyruvate carboxylase of Gram-negative bacteria. As Choi *et al.* (2016) state, "PC is a two-subunit enzyme in a collection of Gram-negative bacteria, with the α subunit containing the BC and the β subunit the CT and BCCP domains" (PMID: 27708276). PycA (BC; α; PP_5347) and PycB (CT + BCCP; β; PP_5346) map directly onto this description. The holoenzyme catalyzes the reaction that Laseke *et al.* (2025)

describe: "Pyruvate carboxylase (PC) catalyzes the carboxylation of pyruvate to oxaloacetate which serves as an important anaplerotic reaction to replenish citric acid cycle intermediates" (PMID: 39725256).

Finding 2 — The reaction catalyzed: the ATP·bicarbonate biotin-carboxylation half-reaction

PycA's specific catalytic contribution is the **first half-reaction** of biotin-dependent carboxylation. The overall pyruvate carboxylase reaction proceeds in two spatially separated half-reactions:

- **Half-reaction 1 (PycA, BC domain):** biotin–[PycB] + HCO_3^- + ATP → carboxybiotin–[PycB] + ADP + P_i. This is the activity annotated for Q88C36 (Rhea RHEA:13501; the biotin-carboxylase sub-activity that operates within EC 6.4.1.1).
- **Half-reaction 2 (PycB, CT domain):** carboxybiotin + pyruvate → oxaloacetate + biotin.
- **Net (EC 6.4.1.1):** pyruvate + HCO_3^- + ATP → oxaloacetate + ADP + P_i.

The BC half-reaction chemistry is well established across the superfamily. Chou & Tong (2011) note that "Biotin carboxylase (BC) activity is shared among biotin-dependent carboxylases and catalyzes the Mg-ATP-dependent carboxylation of biotin using bicarbonate as the CO_2 donor" (PMID: 21592965). Kinetic dissection of the BC domain of *Rhizobium etli* PC by Adina-Zada *et al.* (2012) resolves this half-reaction into "the enzyme-catalyzed bicarbonate-dependent ATP cleavage, carboxylation of biotin, and phosphorylation of ADP by carbamoyl phosphate" (PMID: 22985389) — precisely the chemistry assigned to PycA. Tong (2017) places PycA in context: "Biotin-dependent carboxylases are widely distributed in nature and have central roles in the metabolism of fatty acids, amino acids, carbohydrates, and other compounds" (PMID: 28683917).

Finding 3 — Substrate specificity: PycA carboxylates the biotinyl-lysine of PycB's BCCP domain

The physiological substrate of PycA is **not free biotin** but the biotin prosthetic group covalently tethered to a specific lysine of its partner subunit. PycB (Q88C37) carries the conserved biotin-attachment motif **A-M-K-M**, with the biotinylated residue at **Lys567** within its C-terminal biotin/lipoyl (BCCP) domain (residues ~526–601). This covalently biotinylated lysine is the moiety that PycA carboxylates in half-reaction 1. This mode of substrate presentation —

a mobile, protein-tethered biotin — is the defining feature of biotin-dependent carboxylases and dictates that PycA must physically dock the BCCP arm of PycB into its BC active site to accomplish carboxylation.

Finding 4 — Fully conserved, intact biotin-carboxylase active site (51% identity to *E. coli* AccC)

Direct bioinformatic analysis of Q88C36 confirms a catalytically competent BC enzyme. A Needleman–Wunsch global alignment of Q88C36 (471 aa) against *E. coli* biotin carboxylase AccC (P24182) gives **51.0% identity over 447 aligned positions**. Critically, **all** of the *E. coli* BC catalytic and ATP-binding residues are conserved in Q88C36:

E. coli AccC residue	Q88C36 equivalent	Role
Lys116	Lys115	ATP binding
Lys159	Lys157	ATP binding
His209	His207	catalysis
Glu276	Glu274	catalytic base
Glu288	Glu286	divalent-metal coordination
Asn290	Asn288	divalent-metal coordination

The glycine-rich ATP-binding loop is present (…MLKATSGGGGRGI…) within the ATP-grasp domain (residues ~119–315). These are the very residues identified by Thoden *et al.* for mutagenesis: "Four of these residues of biotin carboxylase, Lys-116, Lys-159, His-209, and Glu-276, were selected for site-directed mutagenesis studies" (PMID: 11346647). Their full conservation in Q88C36 indicates an intact, functional active site. The enzyme requires Mg^{2+} (and can use Mn^{2+}), as for all biotin carboxylases.

Finding 5 — Localization: soluble cytoplasmic protein

PycA carries out its function in the **cytoplasm**. Two lines of evidence support this:

1. **Sequence/topology analysis of Q88C36.** A Kyte–Doolittle hydropathy scan gives a maximum 19-residue window score of **1.26**, below the ~1.6 threshold for a transmembrane helix; the N-terminus (MIKKILIANRGEIA…) shows no cleavable signal peptide. The protein is therefore predicted soluble and cytoplasmic with no membrane insertion or secretion.

2. **Biology of the family.** Bacterial pyruvate carboxylases and biotin carboxylases are cytosolic enzymes. As a comparative example, in the related soil bacterium *Azotobacter vinelandii* the biotin carboxylase (AccC) is a cytoplasmic protein whose ATPase activity is regulated by a cytoplasmic cyclophilin; Dimou *et al.* (2012) report "a physical interaction among AvppiB, encoding the cytoplasmic cyclophilin from the soil nitrogen-fixing bacterium *Azotobacter vinelandii*, and AvaccC, encoding the biotin carboxylase subunit of acetyl-CoA carboxylase" (PMID: 22809506). While that study concerns an ACC biotin carboxylase, it demonstrates the cytoplasmic, soluble nature of bacterial BC subunits generally.

Finding 6 — Mechanistic coupling: swinging biotin and allosteric control by acetyl-CoA

Within the holoenzyme, PycA's active site is coupled to PycB's CT active site by **long-range translocation of the biotinylated carrier domain** — the "swinging arm" mechanism. Liu *et al.* (2018) show that in pyruvate carboxylase "the reaction occurs in two separate catalytic domains, coupled by the long-range translocation of a biotinylated carrier domain (BCCP)" (PMID: 29643369). PycA generates carboxybiotin at its BC site; the BCCP arm of PycB then shuttles the carboxybiotin over tens of ångströms to the CT active site, where pyruvate is carboxylated to oxaloacetate.

This coupling is under **allosteric control by acetyl-CoA**, a hallmark of pyruvate carboxylase. Laseke *et al.* (2025) note that "In most organisms, the PC-catalyzed reaction is allosterically activated by acetyl-coenzyme A" (PMID: 39725256). Mechanistically, acetyl-CoA acts on the carrier translocation step: Liu *et al.* (2018) find that "the allosteric activator, acetyl CoA, promotes one specific intermolecular carrier domain translocation pathway" (PMID: 29643369). The acetyl-CoA binding site is formed largely within the CT/tetramerization (β/PycB-type) region; in *R. etli* PC, key residues include Arg427, Arg429, Arg469, Asp471, and Arg472 (PMID: 27379711; PMID: 26149215; PMID: 22985389). The physiological logic is elegant: high acetyl-CoA signals a need for more oxaloacetate (to condense with acetyl-CoA into citrate, and to sustain the TCA cycle), so acetyl-CoA switches on the anaplerotic PC reaction.

Finding 7 — BC-domain dimerization is required for catalysis

The BC domain of pyruvate carboxylase functions as a dimer, and the dimer interface is catalytically essential. Studies on *Staphylococcus aureus* PC show that "mutations in the dimer interface can produce stable monomers that are still catalytically active," yet several interface mutations (e.g., K442E) and BC-domain chimeras abolish activity, and the isolated *S. aureus* BC domain "is monomeric in solution and catalytically inactive" (PMID: 23286247). *E. coli* BC studies indicate long-range communication between the two active sites of a dimer, and BC shows substrate inhibition by ATP that is competitive against bicarbonate (PMID: 21592965). These properties are expected to apply to PycA given its 51% identity and full active-site conservation.

Finding 8 — Physiological role in *P. putida* KT2440: anaplerosis for TCA/oxaloacetate replenishment

Direct, organism-specific evidence confirms the anaplerotic function. Multi-omics analysis of *P. putida* KT2440 grown on lignin-derived phenolic acids (ferulate, *p*-coumarate, vanillate, 4-hydroxybenzoate) versus succinate showed a large induction of pyruvate carboxylase: Zhou *et al.* (2025) report "Up to 30-fold increase in pyruvate carboxylase and glyoxylate shunt proteins implies a metabolic remodeling" (PMID: 40883435). When *P. putida* grows on aromatic/gluconeogenic substrates that feed carbon into the TCA cycle asymmetrically, it must replenish oxaloacetate to keep the cycle turning and to supply gluconeogenic and amino-acid precursors — exactly the role of pyruvate carboxylase. KEGG maps PP_5347 to the TCA cycle, pyruvate metabolism, carbon fixation, and amino-acid biosynthesis pathways, all consistent with anaplerosis.

Finding 9 — Position within the *P. putida* anaplerotic network

Pyruvate carboxylase (PycA/PycB) operates alongside a set of parallel anaplerotic and gluconeogenic enzymes in *P. putida* KT2440:

- **PEP carboxylase** (*ppc* = PP_1505; K01595): PEP + HCO_3^- → oxaloacetate.
- **NADP-malic enzyme** (*maeB* = PP_5085; K00029): interconverts malate and pyruvate.

Notably, *P. putida* KT2440 has **no single-chain pyruvate carboxylase** (the canonical single-polypeptide KO K01958 is absent from the genome), which is fully consistent with the two-subunit PycA/PycB being the organism's pyruvate carboxylase. The presence of both PEP carboxylase and a two-subunit PC gives *P. putida* redundant routes to oxaloacetate, reflecting the metabolic versatility of this soil bacterium.

Mechanistic Model / Interpretation

The two-subunit pyruvate carboxylase of *P. putida* KT2440

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 Operon:   ... PP_5347 (pycA) --- PP_5346 (pycB) ...
                 |                      |
                 v                      v
           BC subunit (alpha)    CT + BCCP subunit (beta)
           471 aa, Q88C36        602 aa, Q88C37
           ATP-grasp fold        carboxyltransferase + biotin-Lys567
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The holoenzyme executes anaplerosis in two coupled half-reactions:

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HALF-REACTION 1  (PycA, biotin carboxylase active site)
---------------------------------------------------------
   ATP + HCO3-  +  biotin~Lys567-[PycB]
                        |
            Mg2+, BC active site (Lys115, Lys157,
            His207, Glu274, Glu286/Asn288)
                        v
   ADP + Pi     +  carboxy-biotin~Lys567-[PycB]

                |  BCCP "swinging arm" translocates
                |  carboxybiotin ~tens of Angstroms
                v

HALF-REACTION 2  (PycB, carboxyltransferase active site)
---------------------------------------------------------
   carboxy-biotin~Lys567  +  pyruvate
                        v
   biotin~Lys567         +  OXALOACETATE

NET:  pyruvate + HCO3- + ATP  -->  oxaloacetate + ADP + Pi   (EC 6.4.1.1)
Allosteric activator: acetyl-CoA (acts on carrier translocation)
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Why this matters for cellular physiology

Oxaloacetate is consumed continuously — condensed with acetyl-CoA into citrate to run the TCA cycle, and drawn off for gluconeogenesis (via PEP carboxykinase), for aspartate/asparagine and the aspartate-derived amino-acid family, and for pyrimidine biosynthesis. When carbon enters metabolism as pyruvate or as compounds that funnel through pyruvate/acetyl-CoA (including many aromatic and lignin-derived substrates that *P. putida* famously degrades), oxaloacetate would be depleted without a replenishing (anaplerotic) reaction. PycA/PycB pyruvate carboxylase provides exactly this replenishment. The 30-fold induction of PC on aromatic substrates (PMID: 40883435) is the physiological signature of this demand.

The annotation correction, summarized

Attribute	UniProt automated annotation	Corrected assignment (this investigation)
Enzyme complex	Acetyl-CoA carboxylase (ACC)	Two-subunit pyruvate carboxylase (PC)
EC number	6.4.1.2 (implied)	**6.4.1.1**
Product	Malonyl-CoA (fatty-acid initiation)	**Oxaloacetate (anaplerosis)**
Partner subunit	AccB/BCCP + AccA/AccD	**PycB (PP_5346): CT + BCCP**
Genomic neighbor	—	PP_5346 = pyruvate carboxylase β (K01960)
Basis of error	Shared ATP-grasp/BC domain	—

The shared biotin-carboxylase/ATP-grasp domain is common to ACC, PC, propionyl-CoA carboxylase, methylcrotonyl-CoA carboxylase, and urea carboxylase. Automated pipelines keyed on this domain frequently default to the ACC name. The decisive discriminators here are **genomic synteny with a CT+BCCP β-subunit** and the **presence of a separate, complete accBCAD operon** elsewhere in the genome.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
27708276	*A distinct holoenzyme organization for two-subunit pyruvate carboxylase*	Directly describes the α (BC) / β (CT+BCCP) two-subunit PC architecture that matches PycA/PycB in Gram-negative bacteria. **Core support for Finding 1.**
39725256	*Hydrolysis of the acetyl-CoA allosteric activator by S. aureus PC*	Defines the pyruvate→oxaloacetate reaction, its anaplerotic role, and acetyl-CoA allosteric activation. **Supports Findings 1, 2, 6, 8.**
22985389	*Roles of Arg427 and Arg472 in acetyl-CoA binding in PC*	Kinetically dissects the BC-domain half-reaction (bicarbonate-dependent ATP cleavage, biotin carboxylation). **Defines the PycA half-reaction; Findings 2, 6.**
21592965	*Regulation of biotin carboxylase by substrate inhibition and dimerization*	Establishes the Mg-ATP + bicarbonate BC mechanism and substrate inhibition/dimer behavior. **Findings 2, 7.**
11346647	*Site-directed mutagenesis of ATP binding residues of biotin carboxylase*	Identifies the catalytic residues (Lys116, Lys159, His209, Glu276) conserved in Q88C36. **Finding 4.**
28683917	*Striking diversity in holoenzyme architecture of biotin-dependent carboxylases*	Places PycA in the biotin-dependent carboxylase superfamily with a shared BC half-reaction across diverse holoenzymes. **Findings 2, 5.**
29643369	*Allosteric regulation alters carrier domain translocation in PC*	Establishes the swinging-BCCP mechanism coupling BC and CT sites, and acetyl-CoA's effect on translocation. **Finding 6.**
23286247	*Importance of the BC domain dimer for S. aureus PC catalysis*	Shows the BC-domain dimer interface is essential for catalysis; isolated BC domain is inactive. **Finding 7.**
22809506	*Cyclophilin interaction with the biotin carboxylase subunit*	Demonstrates a bacterial BC subunit is cytoplasmic and regulated by protein–protein interaction. **Finding 5.**
40883435	*Quantitative decoding of carbon/energy metabolism*	

PMID	Title (abbrev.)	How it supports the findings
	in P. putida for lignin utilization	Organism-specific: up to 30-fold PC induction during growth on aromatics implies anaplerotic remodeling. **Finding 8.**
27379711 / 26149215	*Allosteric domain residues of R. etli PC*	Map the acetyl-CoA allosteric site (Arg427/Arg429/Arg469/Asp471/Arg472). **Finding 6.**

Note on organism specificity of the evidence: Direct enzymological characterization of *P. putida* KT2440 PycA/PycB has not, to our knowledge, been published. The mechanistic detail above is inferred by strong homology (51% identity of PycA to characterized biotin carboxylases; conserved active site; two-subunit architecture matching Choi *et al.* 2016) and by organism-specific expression data (Zhou *et al.* 2025). This is a well-supported inference, but it remains an inference for the enzymological specifics.

Limitations and Knowledge Gaps

1. **No direct enzymology on *P. putida* PycA/PycB.** Kinetic parameters (kcat, K_m for ATP, bicarbonate, pyruvate), the extent of acetyl-CoA activation, and confirmation of the reaction product for *this specific enzyme* have not been measured. Assignments rest on homology and genomic context.
2. **UniProt name is a known mis-annotation.** The recommended UniProt name ("acetyl-CoA carboxylase biotin carboxylase subunit A") is an automated ARBA/rule-based call driven by the shared ATP-grasp/BC domain and should be treated with caution. The corrected functional assignment (pyruvate carboxylase BC subunit) is supported by orthology and synteny but has not been experimentally validated in KT2440.
3. **Biotinyl-Lys567 of PycB not experimentally confirmed.** The biotinylation site is inferred from the conserved A-M-K-M motif position; direct mass-spectrometric confirmation is lacking.
4. **Structural data are homology-based.** No experimental structure of *P. putida* PycA exists; residue mappings and the hydropathy/topology analysis are computational.
5. **Regulation in vivo.** Whether PycA/PycB is acetyl-CoA-activated in *P. putida* (as in most organisms) and how its expression is controlled at the transcriptional level are not established beyond the proteomic correlation with aromatic-substrate growth.

6. **Interplay with parallel anaplerotic routes.** The relative flux contributions of PC vs. PEP carboxylase (Ppc) vs. malic enzyme under different carbon sources remain to be quantified (e.g., by ^{13}C metabolic flux analysis).

Proposed Follow-up Experiments / Actions

1. **Gene knockout / complementation.** Construct a clean ΔpycA (and ΔpycA ΔpycB) mutant in *P. putida* KT2440 and test growth on pyruvate, lactate, and aromatic/lignin-derived substrates (where anaplerosis via PC is expected to be critical) versus TCA-feeding substrates (succinate) where it should be dispensable. Complement to confirm phenotype specificity.
2. **In vitro enzymology.** Co-express and purify recombinant PycA + PycB; measure the coupled pyruvate → oxaloacetate reaction (malate-dehydrogenase-coupled assay) and the isolated BC half-reaction (ATP/bicarbonate-dependent biotin carboxylation). Determine kcat and K_m values and test acetyl-CoA activation.
3. **Confirm biotinylation site.** Use streptavidin blotting and LC-MS/MS to verify covalent biotin attachment at Lys567 of PycB and demonstrate that PycA carboxylates the holo (biotinylated) but not apo PycB.
4. **Active-site validation.** Site-directed mutagenesis of the conserved catalytic residues predicted here (e.g., Lys115, Glu274) and assay for loss of BC activity, testing the homology-based active-site assignment.
5. **^{13}C metabolic flux analysis (MFA).** Quantify the flux through PC vs. PEP carboxylase vs. malic enzyme across carbon sources (glucose, succinate, ferulate) to define the physiological weighting of PycA/PycB in the *P. putida* anaplerotic network.
6. **Structural determination.** Solve the PycA (and PycA/PycB holoenzyme) structure by cryo-EM or X-ray crystallography to confirm the ATP-grasp fold, the BC-domain dimer interface, and the holoenzyme organization predicted for two-subunit PCs.
7. **Database correction.** Submit the corrected functional assignment (pyruvate carboxylase BC subunit, EC 6.4.1.1) to UniProt with the supporting synteny/orthology evidence to prevent propagation of the ACC mis-annotation.

Conclusion

pycA (PP_5347; Q88C36) encodes the biotin carboxylase (α) subunit of a two-subunit pyruvate carboxylase in *Pseudomonas putida* KT2440. It catalyzes the MgATP- and bicarbonate-dependent carboxylation of the biotin prosthetic group carried on Lys567 of its partner subunit PycB (PP_5346), and, together with PycB's carboxyltransferase activity, converts pyruvate + HCO_3^- + ATP to oxaloacetate + ADP + P_i (EC 6.4.1.1). The enzyme is a soluble, cytoplasmic, anaplerotic protein that replenishes TCA-cycle oxaloacetate — a role of particular importance when *P. putida* grows on aromatic and lignin-derived carbon. The UniProt "acetyl-CoA carboxylase" designation is a domain-driven mis-annotation; *P. putida* KT2440 has a separate, complete acetyl-CoA carboxylase (accBCAD), and PycA is not involved in fatty-acid initiation.
